INCH-POUND
MIL-C-55681/4D
13 July 1994
SUPERSEDING
MIL-C-55681/4C
24 May 1994

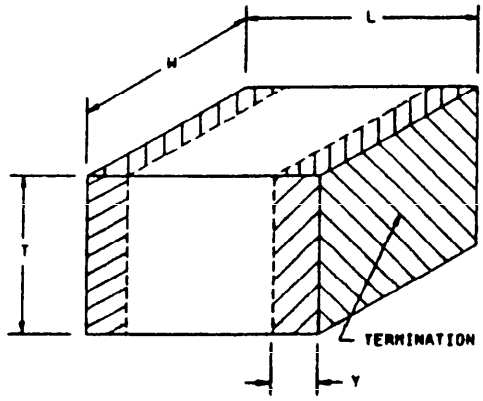
MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CDR11, CDR12, CDR13, AND CDR14 (HIGH FREQUENCY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55681.

Inches	mm
.005	0.13
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.055	1.40
.057	1.45
.102	2.59
.110	2.79



Dimensions					
Style	L	W	T		Y
			Min	Max	
CDR11	.055 ±.015	.055 ±.015	.020	.057	.010 -.005 +.010
CDR12	.055 ±.025	.055 ±.015	.020	.057	.010 -.005 +.010
CDR13	.110 ±.020	.110 ±.020	.030	.102	.015 ±.010
CDR14	.110 -.020, +.035	.110 ±.020	.030	.102	.015 ±.010

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration.

(D) denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multi-layer, unencapsulated, monolithic.

Material: Ceramic.

Terminations:

Styles CDR11 and CDR13: Termination finish M, N, P, or Q.

Styles CDR12 and CDR14: Termination finish S, U, W, Y, or Z. Solder, when applicable, shall have a maximum melting point of 200°C.

Inspection conditions: Solderability examination shall be in accordance with method 208 of MIL-STD-202. All other visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.

Capacitance: See tables I and II.

Capacitance tolerance: See tables I and II.

Rated voltage: See tables I and II.

Operating temperature range: -55°C to +125°C.

Dissipation factor: In accordance with MIL-C-55681.

Insulation resistance:

At +25°C ±5°C: 10^6 megohms, minimum (for 0.1 pF through 470 pF).

10^5 megohms, minimum (for 510 pF through 1,000 pF).

At +125°C ±5°C: 10^5 megohms, minimum (for 0.1 pF through 470 pF).

10^4 megohms, minimum (for 510 pF through 1,000 pF).

Equivalent series resistance: In accordance with MIL-C-55681.

Series resonance: In accordance with MIL-C-55681.

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

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(D) TABLE I. Styles CDR11 and CDR12 capacitor characteristics.

PIN 1/ 2/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pf			V dc
CDR1-B-0R1-B--	0.1	B	BG, BP	50, 150
CDR1-B-0R2-B--	0.2	B	"	"
CDR1-B-0R3----	0.3	B,C	"	"
CDR1-B-0R4----	0.4	B,C	"	"
CDR1-B-0R5----	0.5	B,C,D	"	"
CDR1-B-0R6----	0.6	"	"	"
CDR1-B-0R7----	0.7	"	"	"
CDR1-B-0R8----	0.8	"	"	"
CDR1-B-0R9----	0.9	"	"	"
CDR1-B-1R0----	1.0	"	"	"
CDR1-B-1R1----	1.1	"	"	"
CDR1-B-1R2----	1.2	"	"	"
CDR1-B-1R3----	1.3	"	"	"
CDR1-B-1R4----	1.4	"	"	"
CDR1-B-1R5----	1.5	"	"	"
CDR1-B-1R6----	1.6	"	"	"
CDR1-B-1R7----	1.7	"	"	"
CDR1-B-1R8----	1.8	"	"	"
CDR1-B-1R9----	1.9	"	"	"
CDR1-B-2R0----	2.0	"	"	"
CDR1-B-2R1----	2.1	"	"	"
CDR1-B-2R2----	2.2	"	"	"
CDR1-B-2R4----	2.4	"	"	"
CDR1-B-2R7----	2.7	"	"	"
CDR1-B-3R0----	3.0	"	"	"
CDR1-B-3R3----	3.3	"	"	"
CDR1-B-3R6----	3.6	"	"	"
CDR1-B-3R9----	3.9	"	"	"
CDR1-B-4R3----	4.3	"	"	"
CDR1-B-4R7----	4.7	"	"	"
CDR1-B-5R1----	5.1	"	"	"
CDR1-B-5R6----	5.6	"	"	"
CDR1-B-6R2----	6.2	"	"	"
CDR1-B-6R8----	6.8	B,C,J,K,M	"	"
CDR1-B-7R5----	7.5	"	"	"
CDR1-B-8R2----	8.2	"	"	"
CDR1-B-9R1----	9.1	"	"	"
CDR1-B-100----	10	F,G,J,K,M	"	"
CDR1-B-110----	11	"	"	"
CDR1-B-120----	12	"	"	"
CDR1-B-130----	13	"	"	"
CDR1-B-150----	15	"	"	"
CDR1-B-160----	16	"	"	"
CDR1-B-180----	18	"	"	"
CDR1-B-200----	20	"	"	"
CDR1-B-220----	22	"	"	"
CDR1-B-240----	24	"	"	"
CDR1-B-270----	27	"	"	"
CDR1-B-300----	30	"	"	"
CDR1-B-330----	33	"	"	"
CDR1-B-360----	36	"	"	"
CDR1-B-390----	39	"	"	"
CDR1-B-430----	43	"	"	"

See footnotes at end of table.

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(D) TABLE I. Styles CDR11 and CDR12 capacitor characteristics - Continued.

PIN 1/ 2/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			V dc
CDR1-B-470----	47	F,G,J,K,M	BG, BP	50, 150
CDR1-B-510----	51	"	"	"
CDR1-B-560----	56	"	"	"
CDR1-B-620----	62	"	"	"
CDR1-B-680----	68	"	"	"
CDR1-B-750----	75	"	"	"
CDR1-B-820----	82	"	"	"
CDR1-B-910----	91	"	"	"
CDR1-B-101----	100	"	"	"
CDR1-BP111A---	110	"	BP	50
CDR1-BP121A---	120	"	"	"
CDR1-BP131A---	130	"	"	"
CDR1-BP151A---	150	"	"	"
CDR1-BP161A---	160	"	"	"
CDR1-BP181A---	180	"	"	"
CDR1-BP201A---	200	"	"	"
CDR1-BP221A---	220	"	"	"
CDR1-BP241A---	240	"	"	"
CDR1-BP271A---	270	"	"	"
CDR1-BP301A---	300	"	"	"
CDR1-BP331A---	330	"	"	"
CDR1-BP361A---	360	"	"	"
CDR1-BP391A---	390	"	"	"
CDR1-BP431A---	430	"	"	"
CDR1-BP471A---	470	"	"	"
CDR1-BP511A---	510	"	"	"
CDR1-BP561A---	560	"	"	"
CDR1-BP621A---	620	"	"	"
CDR1-BP681A---	680	"	"	"
CDR1-BP751A---	750	"	"	"
CDR1-BP821A---	820	"	"	"
CDR1-BP911A---	910	"	"	"
CDR1-BP102A---	1000	"	"	"

1/ Complete PIN shall include additional symbols to indicate style, voltage-temperature limits (where applicable), rated voltage (where applicable), capacitance tolerance (where applicable), termination finish ("M", "N", "P", or "Q" for style CDR11, and "S", "U", "W", "Y" or "Z" for style CDR12), and failure rate level.

2/ All capacitors with the 150 volt rating (rated voltage code letter "K") are substitutable, with acquiring agency approval, for the same capacitors which had the 50 volt rating (rated voltage code letter "A") in previous revisions of this document.

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TABLE II. Styles CDR13 and CDR14 capacitor characteristics.

PIN 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			V dc
CDR1-B-0R1-B--	0.1	B	BG, BP	200, 500
CDR1-B-0R2-B--	0.2	B	"	"
CDR1-B-0R3----	0.3	B,C	"	"
CDR1-B-0R4----	0.4	B,C	"	"
CDR1-B-0R5----	0.5	B,C,D	"	"
CDR1-B-0R6----	0.6	"	"	"
CDR1-B-0R7----	0.7	"	"	"
CDR1-B-0R8----	0.8	"	"	"
CDR1-B-0R9----	0.9	"	"	"
CDR1-B-1R0----	1.0	"	"	"
CDR1-B-1R1----	1.1	"	"	"
CDR1-B-1R2----	1.2	"	"	"
CDR1-B-1R3----	1.3	"	"	"
CDR1-B-1R4----	1.4	"	"	"
CDR1-B-1R5----	1.5	"	"	"
CDR1-B-1R6----	1.6	"	"	"
CDR1-B-1R7----	1.7	"	"	"
CDR1-B-1R8----	1.8	"	"	"
CDR1-B-1R9----	1.9	"	"	"
CDR1-B-2R0----	2.0	"	"	"
CDR1-B-2R1----	2.1	"	"	"
CDR1-B-2R2----	2.2	"	"	"
CDR1-B-2R4----	2.4	"	"	"
CDR1-B-2R7----	2.7	"	"	"
CDR1-B-3R0----	3.0	"	"	"
CDR1-B-3R3----	3.3	"	"	"
CDR1-B-3R6----	3.6	"	"	"
CDR1-B-3R9----	3.9	"	"	"
CDR1-B-4R3----	4.3	"	"	"
CDR1-B-4R7----	4.7	"	"	"
CDR1-B-5R1----	5.1	"	"	"
CDR1-B-5R6----	5.6	"	"	"
CDR1-B-6R2----	6.2	"	"	"
CDR1-B-6R8----	6.8	B,C,J,K,M	"	"
CDR1-B-7R5----	7.5	"	"	"
CDR1-B-8R2----	8.2	"	"	"
CDR1-B-9R1----	9.1	"	"	"
CDR1-B-100----	10	F,G,J,K,M	"	"
CDR1-B-110----	11	"	"	"
CDR1-B-120----	12	"	"	"
CDR1-B-130----	13	"	"	"
CDR1-B-150----	15	"	"	"
CDR1-B-160----	16	"	"	"
CDR1-B-180----	18	"	"	"
CDR1-B-200----	20	"	"	"
CDR1-B-220----	22	"	"	"
CDR1-B-240----	24	"	"	"
CDR1-B-270----	27	"	"	"
CDR1-B-300----	30	"	"	"
CDR1-B-330----	33	"	"	"
CDR1-B-360----	36	"	"	"
CDR1-B-390----	39	"	"	"
CDR1-B-430----	43	"	"	"

See footnote at end of table.

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TABLE II. Styles CDR13 and CDR14 capacitor characteristics - Continued.

PIN 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			V dc
CDR1-B-470----	47	F,G,J,K,M	BG, BP	200, 500
CDR1-B-510----	51	"	"	"
CDR1-B-560----	56	"	"	"
CDR1-B-620----	62	"	"	"
CDR1-B-680----	68	"	"	"
CDR1-B-750----	75	"	"	"
CDR1-B-820----	82	"	"	"
CDR1-B-910----	91	"	"	"
CDR1-B-101----	100	"	"	"
CDR1-B-111----	110	"	"	200, 300
CDR1-B-121----	120	"	"	"
CDR1-B-131----	130	"	"	"
CDR1-B-151----	150	"	"	"
CDR1-B-161----	160	"	"	"
CDR1-B-181----	180	"	"	"
CDR1-B-201----	200	"	"	"
CDR1-B-221C---	220	"	"	200
CDR1-B-241C---	240	"	"	"
CDR1-B-271C---	270	"	"	"
CDR1-B-301C---	300	"	"	"
CDR1-B-331C---	330	"	"	"
CDR1-B-361C---	360	"	"	"
CDR1-B-391C---	390	"	"	"
CDR1-B-431C---	430	"	"	"
CDR1-B-471C---	470	"	"	"
CDR1-B-511B---	510	"	"	100
CDR1-B-561B---	560	"	"	"
CDR1-B-621B---	620	"	"	"
CDR1-B-681A---	680	"	"	50
CDR1-B-751A---	750	"	"	"
CDR1-B-821A---	820	"	"	"
CDR1-B-911A---	910	"	"	"
CDR1-B-102A---	1000	"	"	"
CDR1-BP112A---	1100	"	BP	"
CDR1-BP122A---	1200	"	"	"
CDR1-BP132A---	1300	"	"	"
CDR1-BP152A---	1500	"	"	"
CDR1-BP162A---	1600	"	"	"
CDR1-BP182A---	1800	"	"	"
CDR1-BP202A---	2000	"	"	"
CDR1-BP222A---	2200	"	"	"
CDR1-BP242A---	2400	"	"	"
CDR1-BP272A---	2700	"	"	"
CDR1-BP302A---	3000	"	"	"
CDR1-BP332A---	3300	"	"	"
CDR1-BP362A---	3600	"	"	"
CDR1-BP392A---	3900	"	"	"
CDR1-BP432A---	4300	"	"	"
CDR1-BP472A---	4700	"	"	"
CDR1-BP502A---	5000	"	"	"
CDR1-BP512A---	5100	"	"	"

1/ Complete PIN shall include additional symbols to indicate style, voltage-temperature limits (where applicable), rated voltage (where applicable), capacitance tolerance (where applicable), termination finish ("M", "N", "P", or "Q" for style CDR13, and "S", "U", "W", "Y" or "Z" for style CDR14), and failure rate level.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, MC, OS, SH
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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